

Unit 406 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Cable**, 35.1 ac, BAF 20 variable-radius cruise on a 300 ft grid, 17 plots, 205 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	241 sq ft/ac (1 SE of mean, BA 2.3 %)	Type	Sierran mixed conifer
Trees per acre	71	CWHR size / density	5 / D (cover 93 %)
QMD	24.9 in	SDI	297 (50 % of max 598)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	241 sq ft/ac 24.9 in	BA QMD	0 sq ft/ac 0.0 in
TPA	71	TPA	0
Volume	42.0 MBF/ac (8366 cu ft)	Volume	0.0 green tons/ac (0 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~170 sq ft/ac BA	Sampling error (volume)	9.5 % at 95 % (CV 18 %, t = 2.12, n = 17)
Species (BA)	WF 82, PP 60, IC 42, DF 34, SP 22	Stratum standard	40 % of volume - MEETS; plots for standard: 1
		Sampling error (basal area)	4.9 % at 95 % (CV 9 %); secondary, not the standard
		Plot count	17 vs 20-plot local practice, not a handbook standard: SHORT by 3

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). The Keegan ratio is Scribner log scale per cubic foot of delivered sawlog fiber, so applied to total-stem CVTS it overstates sawlog volume; merchantable cubic volume would be the consistent pairing. Sampling error at 95 % confidence with Student's t on plot net cubic volume per acre, the quantity the handbook standards apply to (FSH 2409.12 ch. 40, 41.1(5)(a) and (b)); stratum standard 40 % of volume for tree-measurement sales; the basal-area error is secondary. Stand-summary BA counts every tallied tree (plot count x BAF); the stand and stock tables exclude records flagged by QA, such as an out-of-range DBH, so the two can differ by a few tenths. Sources and URLs: cruise_data.xlsx, Standards and assumptions.

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
8-11.9	1	1.6	1.2	24	0.1
12-15.9	12	12.2	14.1	358	1.8
16-19.9	17	11.5	20.0	577	2.9
20-23.9	41	18.1	48.2	1464	7.3
24-27.9	29	9.6	34.1	1113	5.6
28-31.9	34	8.2	40.0	1409	7.1
32-35.9	21	4.0	24.7	907	4.6
36-39.9	19	2.9	22.4	873	4.4
40-43.9	13	1.6	15.3	615	3.1
44-47.9	7	0.7	8.2	380	1.9
48-51.9	5	0.4	5.9	282	1.4
52-55.9	1	0.1	1.2	65	0.3
56-59.9	3	0.2	3.5	180	0.9
60-63.9	2	0.1	2.4	121	0.6
All	205	71.2	241.2	8366	42.0

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	70	21.1	82.4	26.7	3044	15.3
PP	51	21.5	60.0	22.6	2164	10.9
IC	36	12.1	42.4	25.4	1246	6.3
DF	29	10.2	34.1	24.8	1107	5.6
SP	19	6.3	22.4	25.5	806	4.0
All	205	71.2	241.2	24.9	8366	42.0

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 406: plots on the 300 ft grid (orange = QA flag)

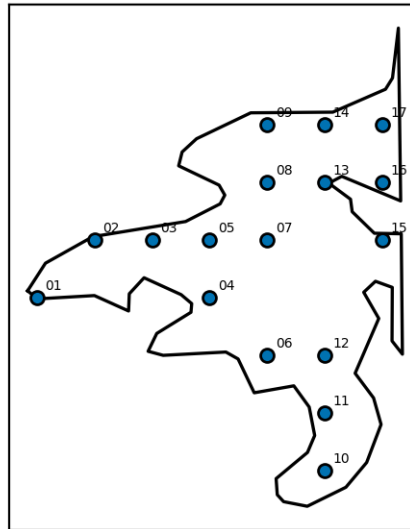


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted; 3 more plots would meet the 20-plot local practice, not a handbook standard

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
406-01	13	260	39	ok
406-02	12	240	45	ok
406-03	12	240	66	ok
406-04	13	260	40	ok
406-05	11	220	59	ok
406-06	14	280	35	ok
406-07	12	240	54	ok
406-08	11	220	67	ok
406-09	10	200	53	ok
406-10	11	220	61	ok
406-11	12	240	45	ok
406-12	11	220	35	ok
406-13	12	240	37	ok
406-14	13	260	48	ok
406-15	11	220	44	ok
406-16	13	260	41	ok
406-17	14	280	44	ok